

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency  $\omega$ : rad/s;  $f$ : Hz

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity  $m/s$ ; *nonmagnetic*:  $\mu_r = 1$

$$u_p = \frac{c}{\sqrt{\epsilon_r}} \qquad c = 3 \times 10^8 \text{ m/s}$$

Wavenumber  $k$  rad/m

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$$

Wavelength & impedance  $m$  &  $\Omega$

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \qquad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \text{ }\Omega$$

Order:  $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$ . Base:  $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$ .

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0 \cos(\omega t - k\hat{k} \cdot \mathbf{r} + \phi_0)\hat{e}$$

$\hat{k}$  = direction of travel (unit vector), e.g.  $+y \Rightarrow \hat{k} = \hat{y}$ .  
 $k$  = wavenumber (rad/m): how fast phase changes in space;  
 $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$ .  
 $\phi_0$  = initial phase constant (rad): shift set by given condition at known  $t$ , position.  
 $\hat{e}$  = polarization direction of  $\mathbf{E}$  (unit vector), e.g.  $E$  along  $x \Rightarrow \hat{e} = \hat{x}$ .  
 $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$ , so  $\hat{k} \cdot \mathbf{r}$  picks the axis, e.g.  $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$ .  
**H** from  $\mathbf{E}$   $\hat{k}$ ,  $\hat{E}$ ,  $\hat{H}$  form right-hand set

$$\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$$

Find  $\phi_0$  given  $E = E_1$  at  $t = 0$ , position  $r_0$

$$E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$$

$\cos^{-1}$  gives  $[0, \pi]$ ; take  $\phi_0$  in that range.

LOSSLESS vs. LOSSY — E & H

|                                            | Lossless ( $\sigma = 0$ )                      | Lossy ( $\sigma \neq 0$ )                        |
|--------------------------------------------|------------------------------------------------|--------------------------------------------------|
| Impedance                                  | $\eta = \sqrt{\mu/\epsilon}$ (real)            | $\eta_c =  \eta_c e^{j\theta_\eta}$ (cmplx)      |
| Prop. factor                               | $e^{-jkz}$                                     | $e^{-\alpha z}e^{-j\beta z}$                     |
| $\hat{\mathbf{H}}$ from $\hat{\mathbf{E}}$ | $\frac{\hat{k} \times \hat{\mathbf{E}}}{\eta}$ | $\frac{\hat{k} \times \hat{\mathbf{E}}}{\eta_c}$ |
| $\hat{\mathbf{E}}$ from $\hat{\mathbf{H}}$ | $-\eta(\hat{k} \times \hat{\mathbf{H}})$       | $-\eta_c(\hat{k} \times \hat{\mathbf{H}})$       |
| E, H phase                                 | in phase                                       | H lags E by $\theta_\eta$                        |

Time domain:  $\mathbf{F}(t) = \text{Re}\{\tilde{\mathbf{F}}e^{j\omega t}\}$  for both  $\mathbf{E}$  and  $\mathbf{H}$ .

DIRECTION OF PROPAGATION

| Arg. contains   | $\hat{k}$  | Arg. contains   | $\hat{k}$  |
|-----------------|------------|-----------------|------------|
| $\omega t - kz$ | $+\hat{z}$ | $\omega t + kz$ | $-\hat{z}$ |
| $\omega t - ky$ | $+\hat{y}$ | $\omega t + ky$ | $-\hat{y}$ |
| $\omega t - kx$ | $+\hat{x}$ | $\omega t + kx$ | $-\hat{x}$ |

Minus before  $k$ -axis  $\Rightarrow$  + direction. Plus  $\Rightarrow$  - direction.

$\hat{k} \times \hat{E}$  — FIND  $\hat{H}$  DIRECTION

| $\hat{k}$  | $\hat{E}$ | $\hat{H}$  | $\hat{k}$  | $\hat{E}$ | $\hat{H}$  |
|------------|-----------|------------|------------|-----------|------------|
| $+\hat{z}$ | $\hat{x}$ | $+\hat{y}$ | $+\hat{z}$ | $\hat{y}$ | $-\hat{x}$ |
| $+\hat{y}$ | $\hat{x}$ | $-\hat{z}$ | $+\hat{y}$ | $\hat{z}$ | $+\hat{x}$ |
| $+\hat{x}$ | $\hat{y}$ | $+\hat{z}$ | $+\hat{x}$ | $\hat{z}$ | $-\hat{y}$ |
| $-\hat{x}$ | $\hat{y}$ | $-\hat{z}$ | $-\hat{x}$ | $\hat{z}$ | $+\hat{y}$ |
| $-\hat{y}$ | $\hat{x}$ | $+\hat{z}$ | $-\hat{z}$ | $\hat{x}$ | $-\hat{y}$ |

Cyclic:  $\hat{x} \times \hat{y} = \hat{z}$ ,  $\hat{y} \times \hat{z} = \hat{x}$ ,  $\hat{z} \times \hat{x} = \hat{y}$ . Reverse  $\Rightarrow$  flip sign.

TRIG PHASE SHIFTS ( $\theta \equiv \omega t - kz$ )

| Input                  | Result         | Input                  | Result         |
|------------------------|----------------|------------------------|----------------|
| $\sin(\theta + \pi/2)$ | $\cos \theta$  | $\cos(\theta + \pi/2)$ | $-\sin \theta$ |
| $\sin(\theta - \pi/2)$ | $-\cos \theta$ | $\cos(\theta - \pi/2)$ | $\sin \theta$  |
| $\sin(\theta \pm \pi)$ | $-\sin \theta$ | $\cos(\theta \pm \pi)$ | $-\cos \theta$ |

POLARIZATION — GENERAL WAVE

pol\_ellipse.jpg

$$\mathbf{E} = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$$

$a_x, a_y$ : bounding box half-widths (V/m);  $\delta$ : phase diff (rad)  
 $\delta = 0$  or  $\pi \Rightarrow$  linear pol. ( $\chi = 0$ , no ellipse)  
Auxiliary angle  $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$   
Rotation angle  $\gamma$  (major axis from  $\hat{x}$ )  $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0) \cos \delta$$

If  $\text{sign}(\gamma) \neq \text{sign}(\delta)$ :  $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$ .  
Ellipticity angle  $\chi$   $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0) \sin \delta$$

Axial ratio  $R = 1/|\tan \chi|$  major axis =  $R \times$  minor;  $R \geq 1$

$\chi$  SHAPE & HANDEDNESS

| $\chi$ describes the shape of the ellipse and rotation direction. |                   |            |
|-------------------------------------------------------------------|-------------------|------------|
| $\chi$ (rad)                                                      | Polarization Type | Handedness |
| 0                                                                 | Linear            | None       |
| $(0, \pi/4)$                                                      | RHEP              | RH (CCW)   |
| $\pi/4$                                                           | RHC               | RH (CCW)   |
| $-\pi/4$                                                          | LHC               | LH (CW)    |
| $(-\pi/4, 0)$                                                     | LHEP              | LH (CW)    |

QUADRANT RULE FOR  $\tan(2\gamma)$

$$N = \sin(2\psi_0) \cos \delta, \quad D = \cos(2\psi_0):$$

| $N$ | $D$ | $2\gamma$ (deg)            | $2\gamma$ (rad)       |
|-----|-----|----------------------------|-----------------------|
| +   | +   | $\arctan(N/D)$             | $\arctan(N/D)$        |
| +   | -   | $180^\circ - \arctan N/D $ | $\pi - \arctan N/D $  |
| -   | -   | $180^\circ + \arctan N/D $ | $\pi + \arctan N/D $  |
| -   | +   | $360^\circ - \arctan N/D $ | $2\pi - \arctan N/D $ |

$\gamma = (2\gamma)/2$ .  $D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$ .  
Convert:  $\theta_{\text{rad}} = \theta_o \times \frac{\pi}{180}$ ;  $\theta_o = \theta_{\text{rad}} \times \frac{180}{\pi}$ .

PHASOR  $\leftrightarrow$  TIME DOMAIN

| Phasor coeff | Time domain (add $e^{-\alpha z}$ )      |
|--------------|-----------------------------------------|
| $A$ (real)   | $A \cos(\omega t - \beta z)$            |
| $-jA$        | $A \sin(\omega t - \beta z)$            |
| $+jA$        | $-A \sin(\omega t - \beta z)$           |
| $Ae^{j\phi}$ | $A \cos(\omega t - \beta z + \phi)$     |
| $A + jB$     | $ Z  \cos(\omega t - \beta z + \theta)$ |

$|Z| = \sqrt{A^2 + B^2}$ ,  $\theta = \arctan(B/A)$  (check quadrant)

AVERAGE POWER DENSITY (POYNTING)

General  $W/m^2$

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$$

Lossless shortcut  $|E_0|^2 = a_x^2 + a_y^2 + a_z^2$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta} \hat{k}$$

Direction =  $\hat{k}$ . If wave goes in  $-\hat{x}$ , then  $\mathbf{S}_{\text{av}}$  is a negative  $x$ -component.

LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon_r\epsilon_0}$$

| $\epsilon''/\epsilon'$ | Type                | Use       |
|------------------------|---------------------|-----------|
| $> 100$                | Good conductor      | GC approx |
| $0.01$ to $100$        | Quasi-conductor     | Exact eqs |
| $< 0.01$               | Low-loss dielectric | LD approx |

$\epsilon' = \epsilon_r\epsilon_0$ ;  $\epsilon'' = \sigma/\omega$ ; nonmag.:  $\mu = \mu_0$ ; free space:  $\epsilon_r = \mu_r = 1$ .

ANY MEDIUM — EXACT ( $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$ )

$$\alpha = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X-1} \text{ Np/m}, \quad \beta = \omega\sqrt{\frac{\mu\epsilon'}{2}}\sqrt{X+1} \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f} \text{ (m)}, \quad u_p = \frac{\omega}{\beta} \text{ (m/s)}$$

$$\eta_c = \frac{\eta}{X^{1/2}}e^{j\theta_\eta} \text{ (}\Omega\text{)}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'} \text{ (rad)}$$

Rect:  $\eta_c = |\eta_c| \cos \theta_\eta + j|\eta_c| \sin \theta_\eta$ ,  $|\eta_c| = \eta/X^{1/2}$   
 $\eta = \eta_0/\sqrt{\epsilon_r}$ ;  $\epsilon' = \epsilon_r\epsilon_0$ .

LOSSLESS ( $\sigma = 0$ )

$$\alpha = 0 \quad \beta = \omega\sqrt{\mu\epsilon} = k \text{ (rad/m)} \quad \lambda = u_p/f \text{ (m)}$$

$$\eta_c = \eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{, real; } \theta_\eta = 0)$$

$$u_p = 1/\sqrt{\mu\epsilon} = c/\sqrt{\epsilon_r} \text{ (m/s)}$$

LOW-LOSS MEDIUM ( $\epsilon''/\epsilon' \ll 1$ )

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma\eta}{2} \text{ (Np/m)}$$

$$\beta \approx \omega\sqrt{\mu\epsilon'} = \frac{\omega\sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad u_p \approx \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s)}$$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} = \eta \text{ (}\Omega\text{, } \angle \approx 0)$$

GOOD CONDUCTOR ( $\epsilon''/\epsilon' \gg 1$ )

$$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma} \text{ (Np/m, rad/m)}$$

$$u_p \approx \sqrt{4\pi f / \mu \sigma} \text{ (m/s)}, \quad \lambda \approx u_p/f \text{ (m)}$$

$$\eta_c \approx (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma} \text{ (}\Omega\text{)}$$

$\hat{\mathbf{H}}$  PHASOR  $\rightarrow \hat{\mathbf{E}}$  IN LOSSY MEDIA

Step 1 — read  $\alpha, \beta$

$$\tilde{\mathbf{H}} \propto e^{-\alpha y}e^{-j\beta y} \Rightarrow \gamma = \alpha + j\beta$$

Step 2 —  $\eta_c$  (nonmagnetic:  $\mu_r = 1$ )

$$\eta_c = \frac{j\omega\mu_0}{\gamma} = \frac{\omega\mu_0(\beta + j\alpha)}{\alpha^2 + \beta^2}$$

Step 3 — find  $\tilde{\mathbf{E}}$

$$\tilde{\mathbf{E}} = -\eta_c (\hat{k} \times \tilde{\mathbf{H}})$$

Step 4 — time domain

$$Ae^{j\phi}e^{-\alpha y} \xrightarrow{\text{Re}\{ \cdot e^{j\omega t} \}} Ae^{-\alpha y} \cos(\omega t - \beta y + \phi)$$

Convert each component  $A + jB \rightarrow |Z|\angle\theta$  first.

WAVE PARAMS

$\omega = 2\pi f$  — ang. freq (rad/s)  
 $k = \omega\sqrt{\epsilon_r}/c$  — wavenumber (rad/m)  
 $\lambda = 2\pi/k$  — wavelength (m)  
 $u_p = c/\sqrt{\epsilon_r}$  — phase vel. (m/s)  
 $\eta = \eta_0/\sqrt{\epsilon_r}$  — impedance ( $\Omega$ )  
 $c = 3 \times 10^8$  m/s  
 $\eta_0 = 377 \Omega$  (free space)

LOSSLESS MEDIA

$\sigma = 0$  (vacuum, air, ideal dielectric)  
 $\epsilon'' = \sigma/\omega = 0 \Rightarrow \epsilon = \epsilon'$  (real)  
 $\alpha = 0$ ; no attenuation  
 $\eta = \eta_0/\sqrt{\epsilon_r}$  real ( $\Omega$ )  
 $\eta_0 = 377 \Omega$  (free space)  
**E** and **H** in phase ( $\theta_\eta = 0$ )

LOSSY MEDIA

$\alpha$  — atten. const (Np/m)  
 $\beta$  — phase const (rad/m)  
 $\gamma = \alpha + j\beta$  — prop. const ( $\text{m}^{-1}$ )  
 $\eta_c$  — complex impedance ( $\Omega$ )  
 $\epsilon' = \epsilon_r\epsilon_0$ ;  $\epsilon'' = \sigma/\omega$  (F/m)  
 $\mu_r$  — rel. permeability ( $\mu_r = 1$ : nonmagnetic)  
 $\mu = \mu_r\mu_0$ ;  $\mu_0 = 4\pi \times 10^{-7}$  H/m  
 $\delta_s = 1/\alpha$  — skin depth (m)

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$   
 $\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$   
 $a_R, a_L, a_x$  in V/m;  $k$  in rad/m  
Linear:  $a_R = a_L = a_x/2$   
 $-j = e^{-j\pi/2}$ :  $\cos \rightarrow \sin$   
 $+j = e^{+j\pi/2}$ :  $\cos \rightarrow -\sin$

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$ ;  $\sqrt{9} = 3$   
 $\sqrt{2.5} \approx 1.581$ ;  $\sqrt{2} \approx 1.414$   
 $\arccos(2/3) \approx 48.2^\circ = 0.84$  rad  
 $\arctan(4/3) \approx 53.1^\circ = 0.927$  rad  
 $\arctan(3/4) \approx 36.9^\circ = 0.644$  rad  
 $\arctan(24/7) \approx 73.7^\circ = 1.287$  rad